Formative Assessment 4

```r
suppressPackageStartupMessages({
  library(readxl)
  library(tidyverse)
  library(ggplot2)
  library(car)
  library(PerformanceAnalytics)
  library(xts)
  library(rugarch)
  library(quantmod)
  library(tseries)
  library(forecast)
})
```

Stock Returns

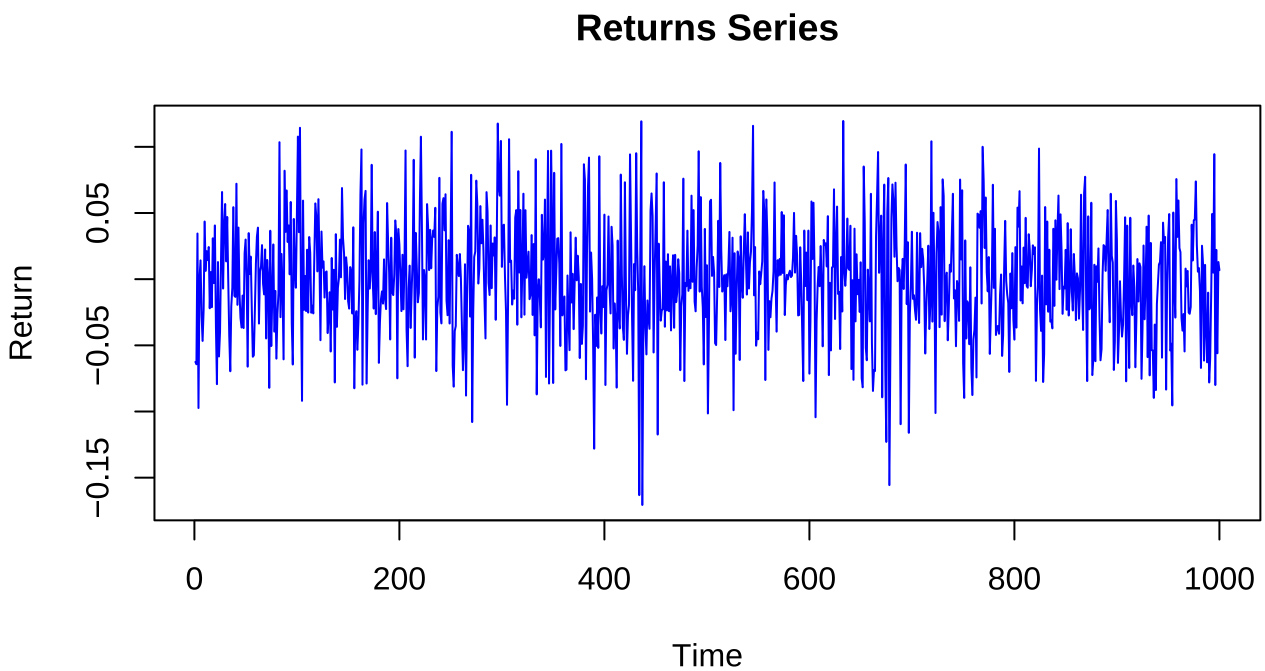
```r
stock <- read_csv('dataset_stock_returns.csv', show_col_types = FALSE)
head(stock)
```

```
## # A tibble: 6 x 1
##     returns
##       <dbl>
## 1 -0.0625
## 2 -0.0643
## 3  0.0345
## 4 -0.0974
## 5  0.000708
## 6  0.0142
```

Visual Exploration of Volatility

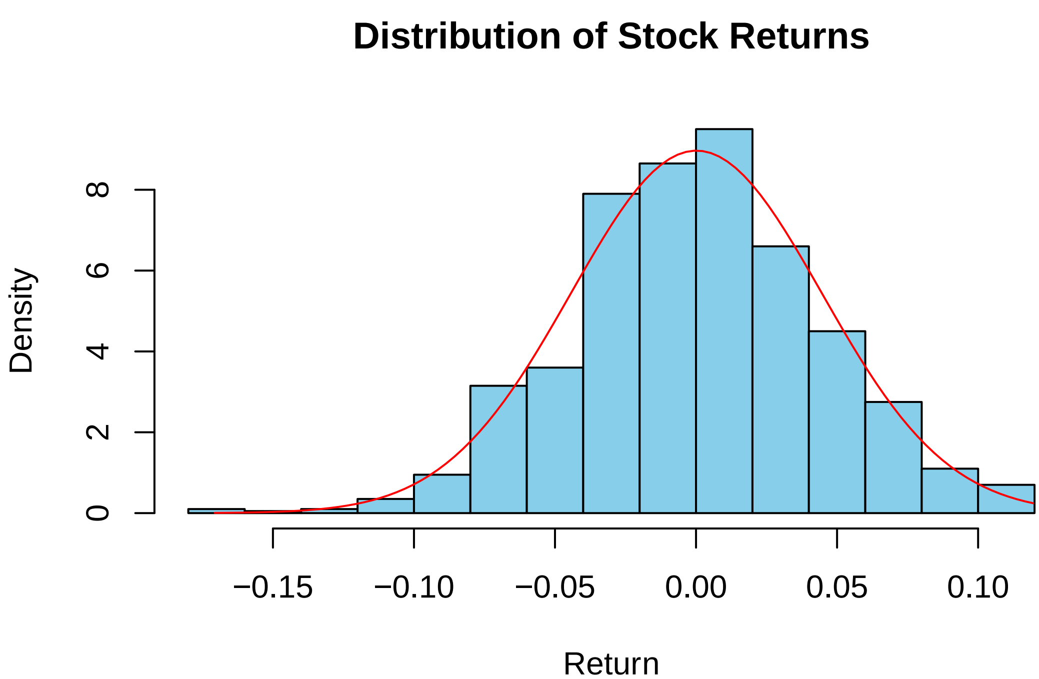
Return Series and Rolling Standard Deviation

```r
plot(1:nrow(stock), stock$returns, type = "l", col = "blue",
     main = "Returns Series", xlab = "Time", ylab = "Return")
```



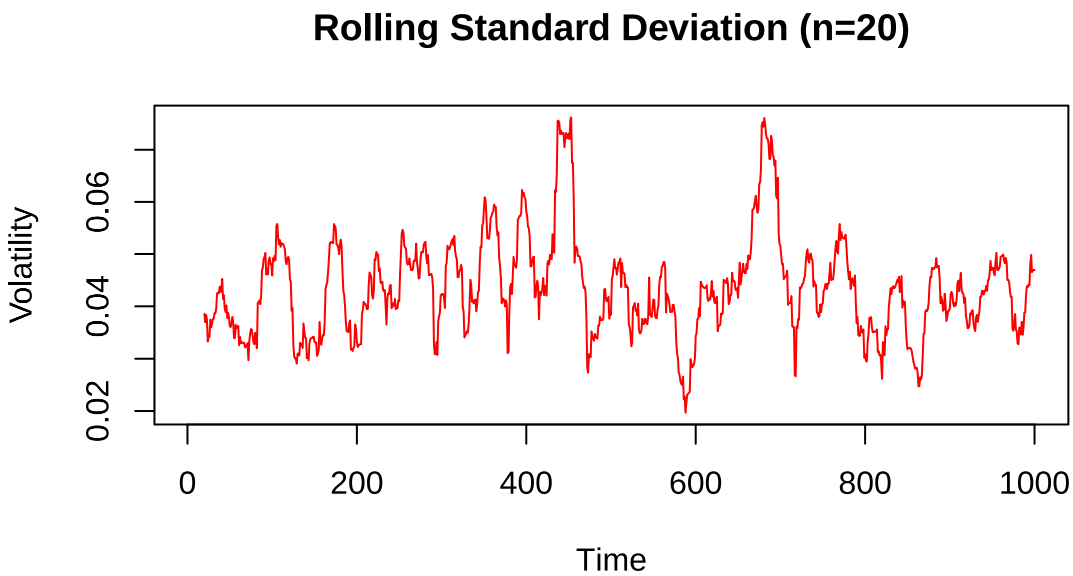
The return series plot shows returns fluctuating randomly around 0, showing alternating periods of high and low variability. Notably, two distinct time intervals stand out where returns exhibit large negative and positive movements, preceded and followed by periods of relatively small changes. This pattern is indicative of volatility clustering.

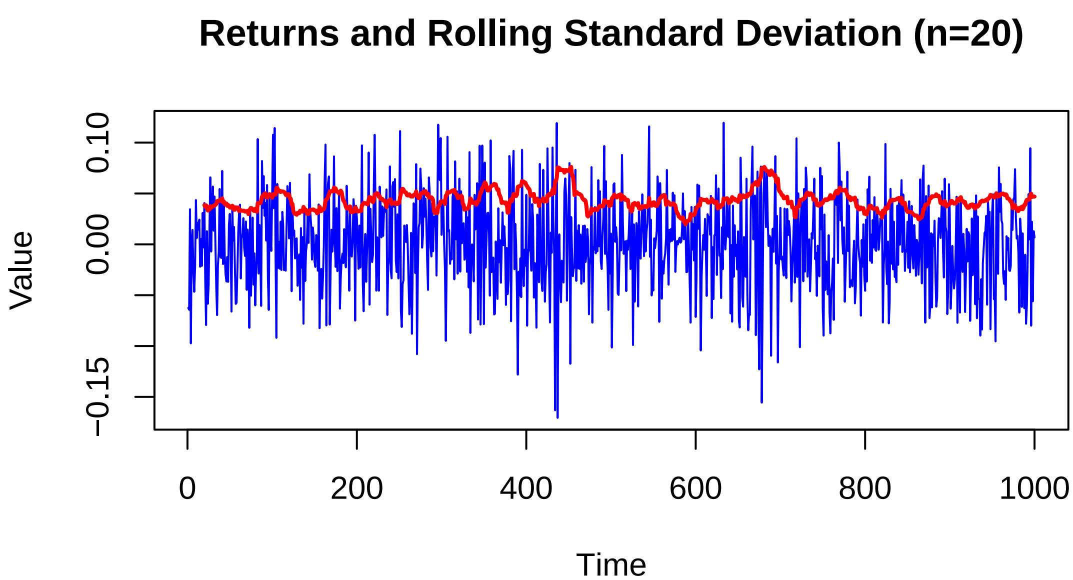
```
hist(stock$returns, main="Distribution of Stock Returns",
     col = "skyblue", xlab= "Return", freq=FALSE)

x <- seq(min(stock$returns, na.rm = TRUE),
         max(stock$returns, na.rm = TRUE),
         length = 100)

lines(x,dnorm(x,mean = mean(stock$returns, na.rm = TRUE),
              sd = sd(stock$returns, na.rm = TRUE)), col="red")
```



The plot of returns is not symmetrically distributed, rather it is negatively skewed. This indicates that on average, returns are negative. Compared to the normal distribution, the returns have moderate kurtosis, meaning that extreme changes happen moderately.

```
par(mfrow=c(2,1))
rolling_sd <- runSD(stock$returns, n = 20, cumulative = FALSE)
plot(1:nrow(stock), rolling_sd, type = "l", col = "red",
     main = "Rolling Standard Deviation (n=20)", xlab = "Time", ylab = "Volatility")
plot(1:nrow(stock), stock$returns, type = "l", col = "blue",
     main = "Returns and Rolling Standard Deviation (n=20)",
     xlab = "Time", ylab = "Value")
lines(1:nrow(stock), rolling_sd, col = "red", lwd = 2)
```

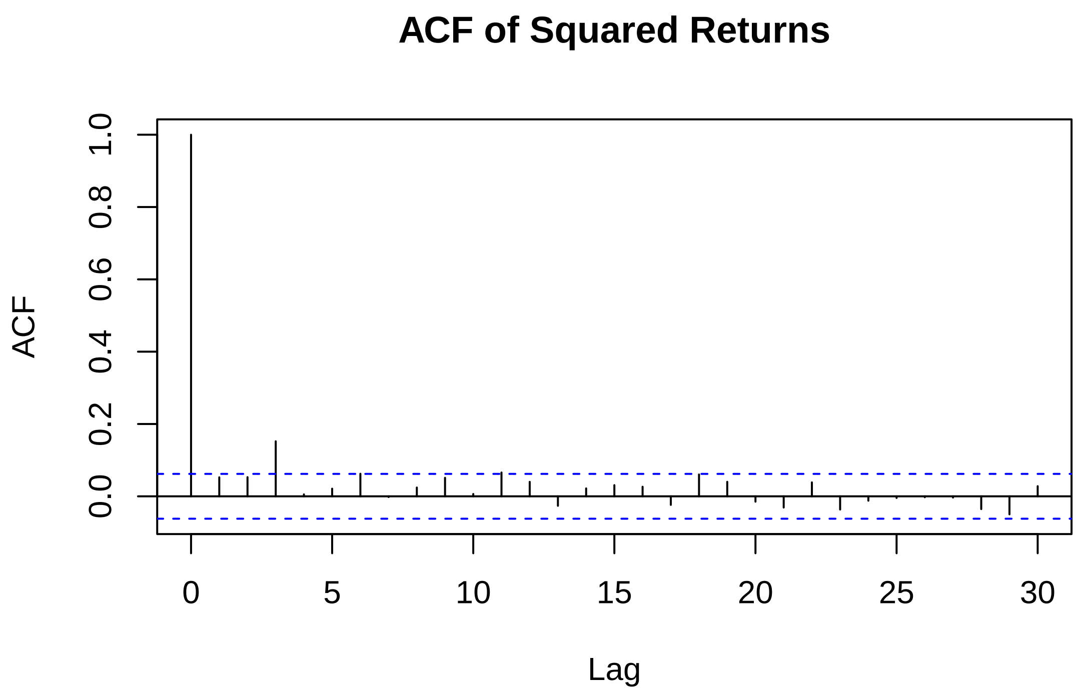




Assuming 20 trading days per month, the rolling standard deviation line shows the monthly volatility of the stock returns. It rises sharply during turbulent periods, reflecting increased volatility, and falls during calm periods when the market stabilizes. This behavior suggests that the variance of returns is not constant over time, implying conditional heteroskedasticity.

Plot ACF of Squared Returns

```
acf(stock^2, main = "ACF of Squared Returns")
```



The ACF of squared returns drops quickly to zero and is insignificant at lags 1–2 but becomes significant again at lag 3. This pattern suggests an absence of immediate volatility clustering, but the presence of some delayed or periodic dependence in volatility roughly every three periods. The changing variance of returns over time indicates conditional heteroskedasticity, consistent with the presence of volatility dynamics. This could point to a potential ARCH(3)-type effect, where volatility depends on shocks occurring three periods earlier.

Model Estimation

Model 1: ARCH(3)

ARCH(q) model is a special case of GARCH where the conditional variance depends only on past squared shocks (no lagged variance term).

```
arch3_spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(0,3)),
  mean.model     = list(armaOrder = c(0,0), include.mean = TRUE),
  distribution.model = "std")

arch3_fit <- ugarchfit(spec = arch3_spec, data = stock$returns)
```

Coefficients and Standard Errors

```
arch3_fit@fit$matcoef
```

```
##           Estimate   Std. Error      t value     Pr(>|t|)
## mu    2.166247e-04 1.402337e-03    0.1544741 0.8772359606
## omega 2.006030e-06 5.356010e-07    3.7453811 0.0001801201
## beta1 8.228864e-02 3.618336e-03   22.7421199 0.0000000000
```

```
## beta2 1.602303e-01 3.679859e-03   43.5424966 0.0000000000
## beta3 7.564807e-01 1.798199e-04 4206.8807577 0.0000000000
## shape 2.884946e+01 1.656297e+01    1.7418045 0.0815426586
```

AIC

```r
infocriteria(arch3_fit)[1]
```

```
## [1] -3.378067
```

```r
(-2*likelihood(arch3_fit))/nrow(stock)+2*(length(arch3_fit@fit$coef))/nrow(stock)
```

```
## [1] -3.378067
```

Model 2: GARCH(1,3)

```r
garch13_spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 3)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"
)

garch13_fit <- ugarchfit(spec = garch13_spec, data = stock$returns)
```

Coefficient Estimates and Standard Errors

```r
garch13_fit@fit$matcoef
```

```
##            Estimate   Std. Error      t value   Pr(>|t|)
## mu     1.364974e-04 1.374164e-03 0.0993312051 0.92087530
## omega  2.055112e-04 1.227384e-04 1.6743846933 0.09405502
## alpha1 5.314097e-02 3.481535e-02 1.5263660840 0.12691872
## beta1  8.422449e-01 1.268361e+00 0.6640420230 0.50666342
## beta2  2.433679e-04 1.391115e+00 0.0001749445 0.99986041
## beta3  8.279368e-05 9.945168e-02 0.0008325016 0.99933576
## shape  8.124796e+01 1.455987e+02 0.5580265746 0.57682624
```

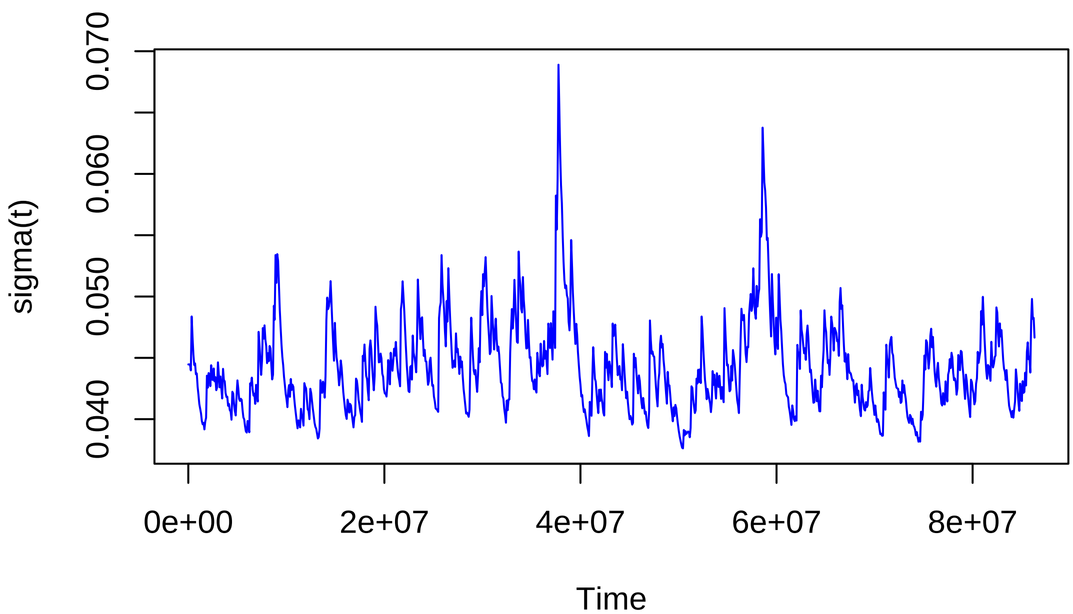
```r
infocriteria(garch13_fit)
```

```
##
## Akaike       -3.389261
## Bayes        -3.354907
## Shibata      -3.389358
## Hannan-Quinn -3.376204
```

Diagnostics and Volatility Interpretation

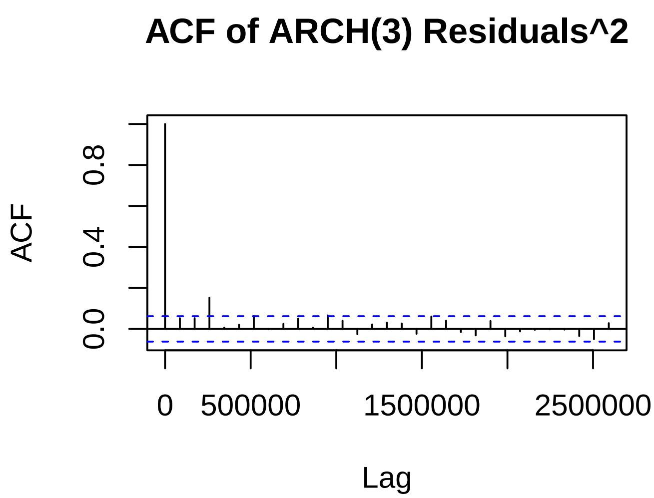
Plot conditional volatility and compare across models.

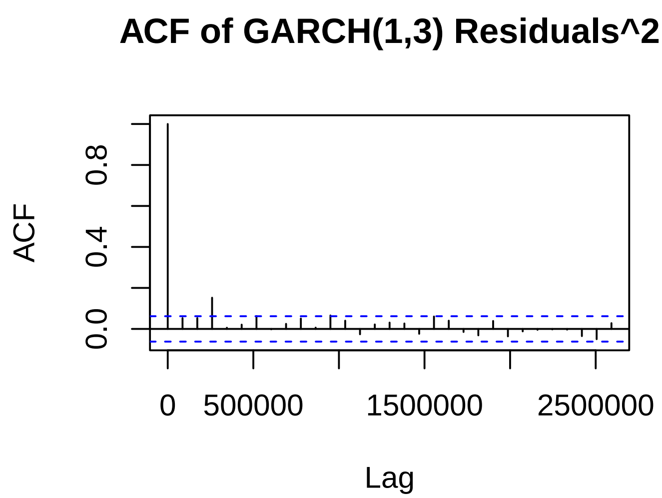
```r
plot.ts(sigma(garch13_fit), ylab="sigma(t)", col="blue")
```



Analyze residuals for remaining ARCH effects (using ACF of residuals squared or ARCH-LM test).

```r
par(mfrow=c(1,2), mai = c(1, 0.8, 0.7, 0.3))
acf(residuals(arch3_fit)^2, main ="ACF of ARCH(3) Residuals^2")
acf(residuals(garch13_fit)^2, main ="ACF of GARCH(1,3) Residuals^2")
```





Comment on which model better captures volatility persistence and asymmetry.

Visual inspection of the returns and rolling standard deviation plots shows clear volatility clustering: turbulent periods are followed by sustained high volatility, and calm periods are followed by extended low volatility. The rolling SD line rises and decays gradually rather than abruptly, suggesting persistent volatility. This pattern supports a GARCH(1,1) model, which accounts for both recent shocks (ARCH term) and persistent effects from past volatility (GARCH term), making it more appropriate than a simple ARCH model.

Volatility Forecasting

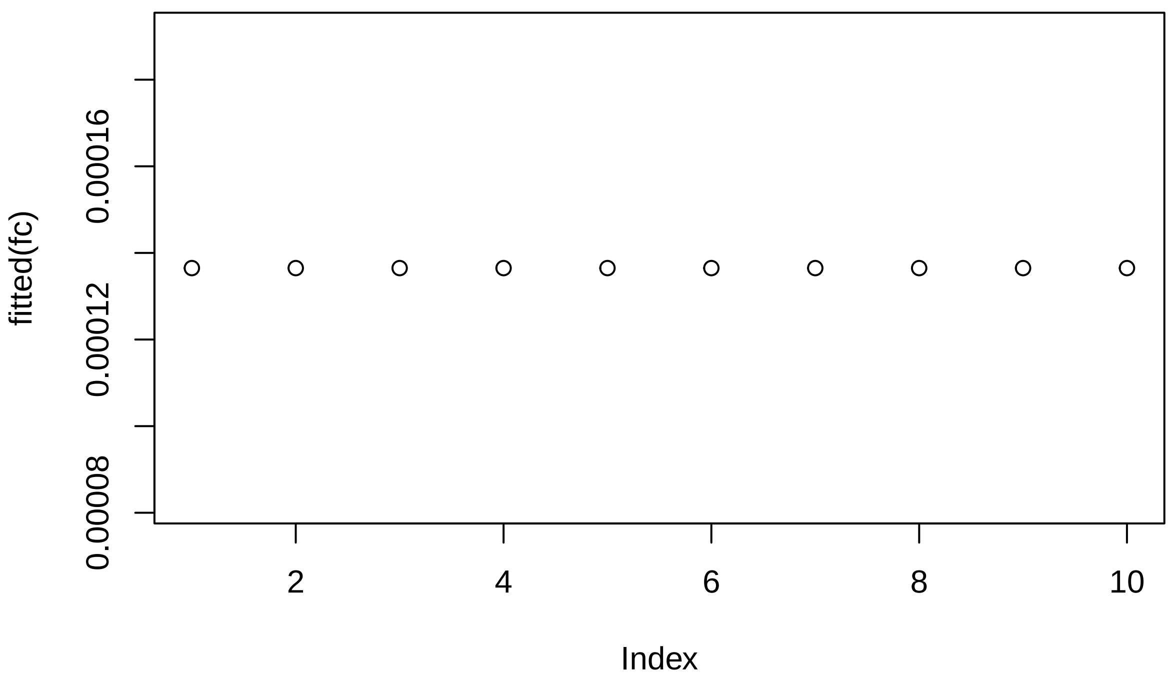
Forecast conditional volatility for 10 future periods

```r
fc <- ugarchforecast(fitORspec =garch13_fit, n.ahead=10)
fc
```

```
##
## *------------------------------------*
## *       GARCH Model Forecast         *
## *------------------------------------*
## Model: sGARCH
## Horizon: 10
## Roll Steps: 0
## Out of Sample: 0
##
## 0-roll forecast [T0=1972-09-27]:
##         Series   Sigma
## T+1  0.0001365 0.04518
## T+2  0.0001365 0.04510
## T+3  0.0001365 0.04503
## T+4  0.0001365 0.04496
## T+5  0.0001365 0.04490
## T+6  0.0001365 0.04485
## T+7  0.0001365 0.04480
## T+8  0.0001365 0.04476
## T+9  0.0001365 0.04472
## T+10 0.0001365 0.04469
```

forecasted volatility path with confidence intervals

```r
plot(fitted(fc))
```



what happens to volatility over the forecast horizon (does it decay, persist, or show asymmetry?)

Reflection and Application

Discuss in which contexts (finance, energy, currency) volatility modeling is most relevant.

Identify any limitations or possible model improvements.

Bitcoin Returns

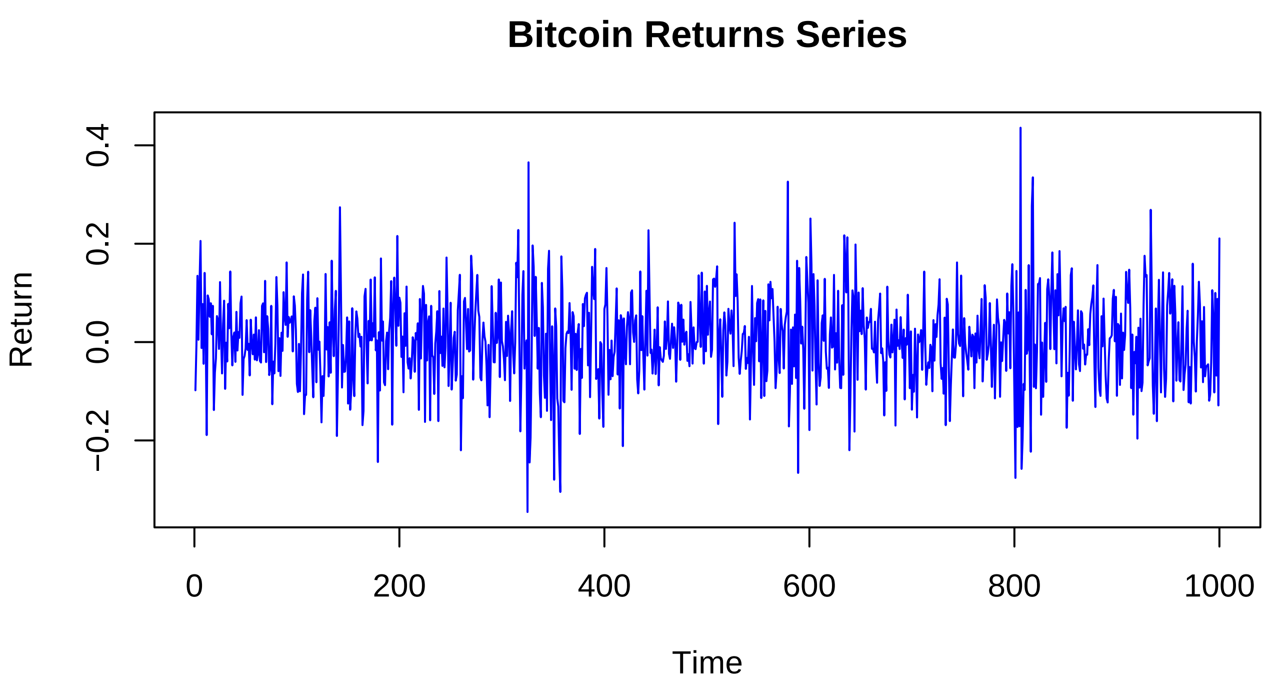
```r
bitcoin <- read_csv('dataset_bitcoin_returns.csv', show_col_types = FALSE)
head(bitcoin)
```

```
## # A tibble: 6 x 1
##    returns
##      <dbl>
## 1 -0.0980
## 2 -0.00617
## 3  0.135
## 4  0.00469
## 5  0.119
## 6  0.206
```

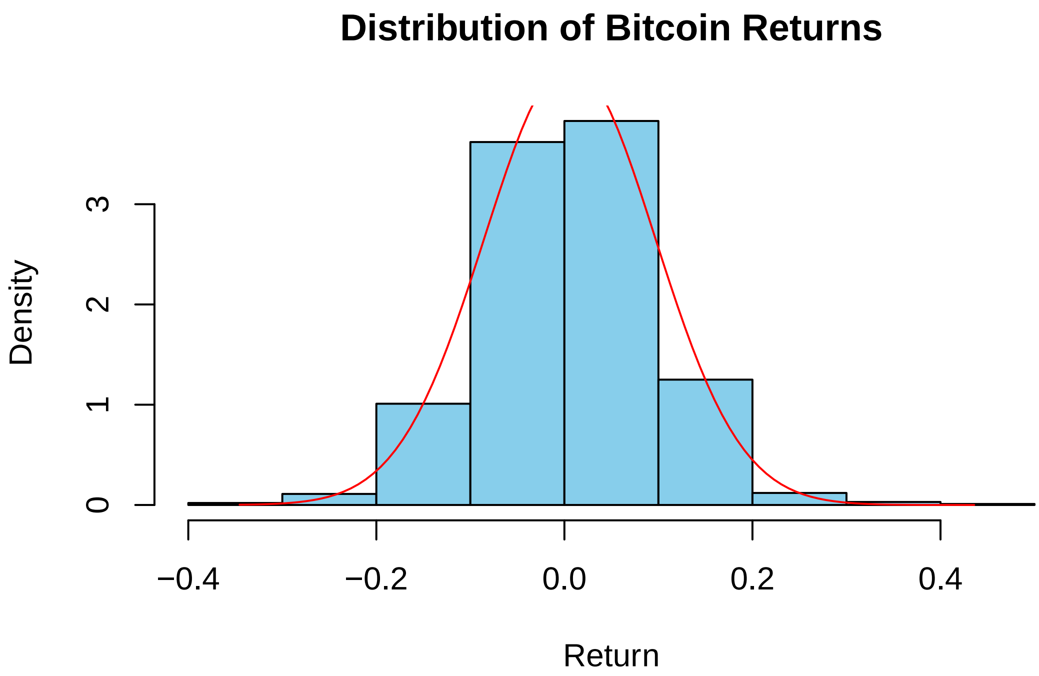
Visual Exploration of Volatility

Return Series and Rolling Standard Deviation

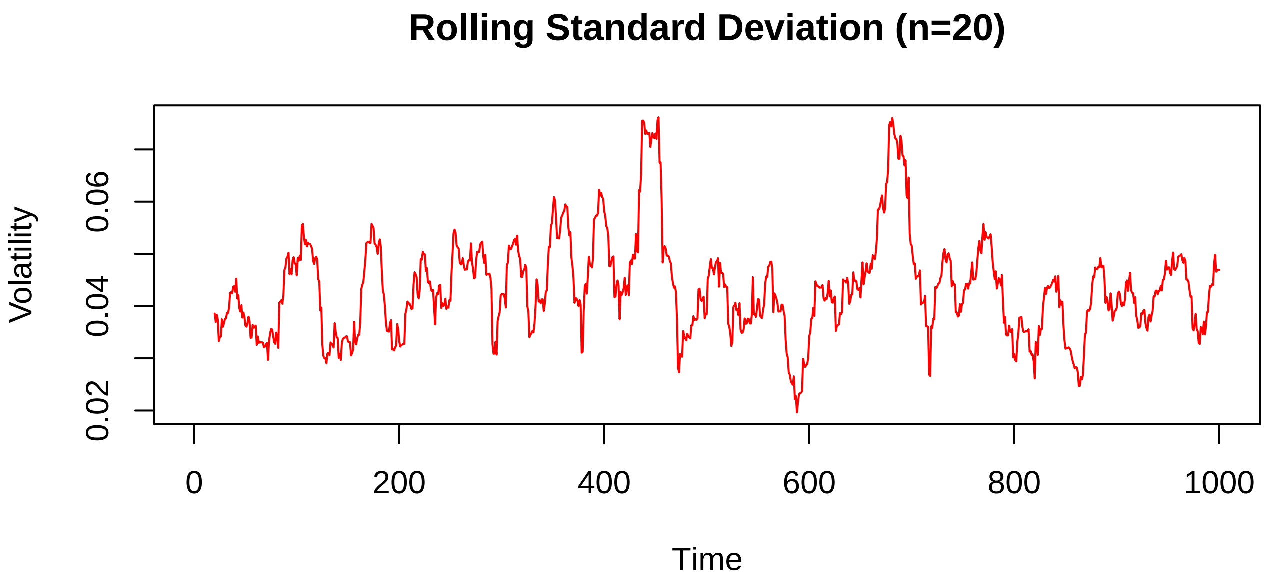
```r
plot(1:nrow(bitcoin), bitcoin$returns, type = "l", col = "blue",
     main = "Bitcoin Returns Series", xlab = "Time", ylab = "Return")
```

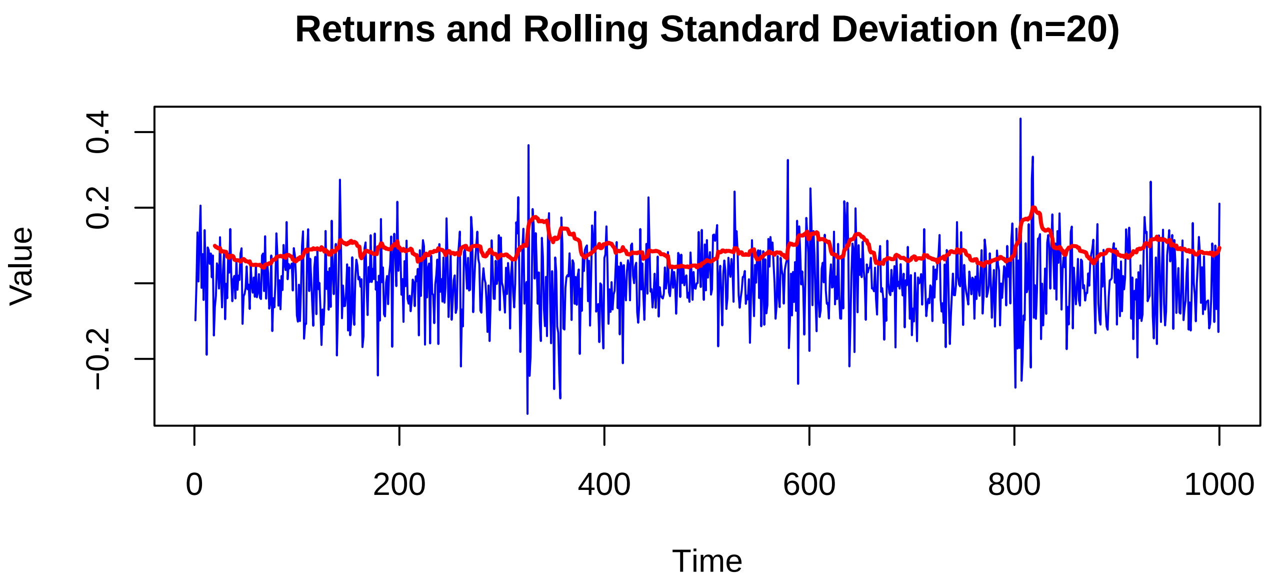



```r
hist(bitcoin$returns, main="Distribution of Bitcoin Returns",
     col = "skyblue", xlab= "Return", freq = FALSE)

x <- seq(min(bitcoin$returns, na.rm = TRUE),
         max(bitcoin$returns, na.rm = TRUE),
         length = 100)

lines(x,dnorm(x,mean = mean(bitcoin$returns, na.rm = TRUE),
              sd = sd(bitcoin$returns, na.rm = TRUE)), col="red")
```

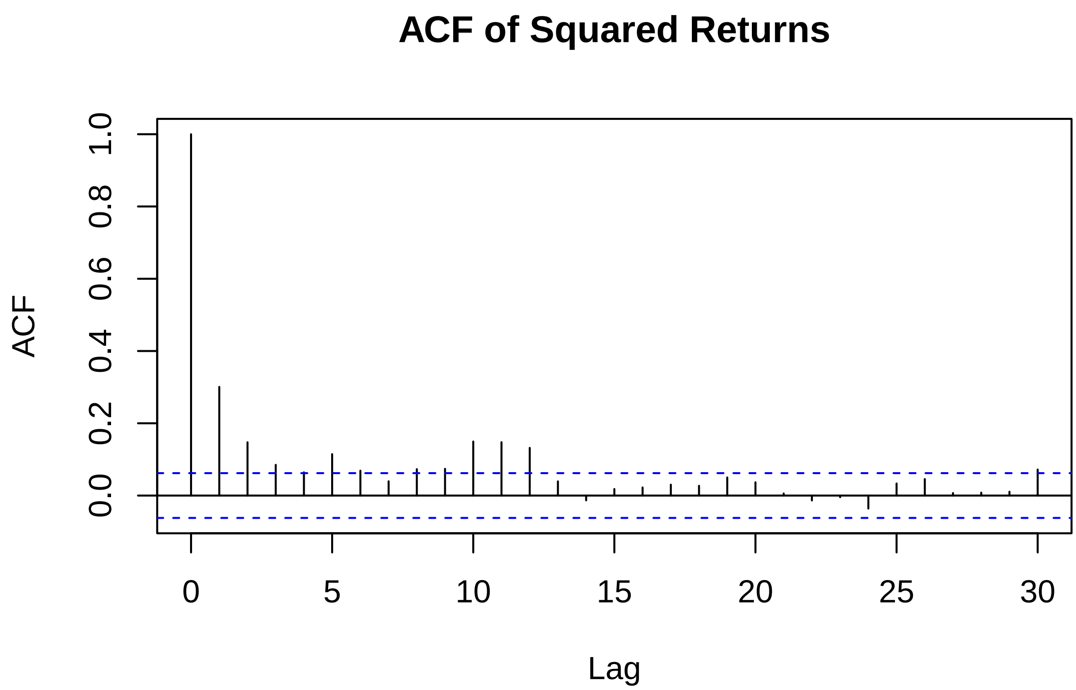



```
par(mfrow=c(2,1))
plot(1:nrow(bitcoin), rolling_sd, type = "l", col = "red",
     main = "Rolling Standard Deviation (n=20)", xlab = "Time", ylab = "Volatility")
rolling_sd <- runSD(bitcoin$returns, n = 20, cumulative = FALSE)
plot(1:nrow(bitcoin), bitcoin$returns, type = "l", col = "blue",
     main = "Returns and Rolling Standard Deviation (n=20)",
     xlab = "Time", ylab = "Value")
lines(1:nrow(bitcoin), rolling_sd, col = "red", lwd = 2)
```






```
acf(bitcoin^2, main = "ACF of Squared Returns")
```



Slow decay, pattern

Model Estimation

Model 1 GARCH(1,1)

```
garch11_spec <-  ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "norm"
)

garch11_fit <- ugarchfit(garch11_spec, data = bitcoin$returns)
```

```
garch11_fit@fit$matcoef
```

```
##             Estimate   Std. Error   t value     Pr(>|t|)
## mu     0.0047440716 0.0026305603  1.803445 7.131831e-02
## omega  0.0005469361 0.0002363475  2.314119 2.066121e-02
## alpha1 0.0942331023 0.0226437421  4.161552 3.160925e-05
## beta1  0.8390589415 0.0442503382 18.961639 0.000000e+00
```

```
infocriteria(garch11_fit)
```

```
##
## Akaike       -2.026118
## Bayes        -2.006487
## Shibata      -2.026150
## Hannan-Quinn -2.018657
```

Model 2

Record estimated parameters, standard errors, and information criteria (AIC, BIC).

Diagnostics and Volatility Interpretation

Plot conditional volatility and compare across models.

Analyze residuals for remaining ARCH effects (using ACF of residuals squared or ARCH-LM test).

Comment on which model better captures volatility persistence and asymmetry.

Volatility Forecasting

Forecast conditional volatility for 10 future periods

forecasted volatility path with confidence intervals

what happens to volatility over the forecast horizon (does it decay, persist, or show asymmetry?)

Reflection and Application

Discuss in which contexts (finance, energy, currency) volatility modeling is most relevant.

Identify any limitations or possible model improvements.